The Cosmic Web as a Generalized Murray Network:

Transport-Optimal Branching Under Gravitational–Dark Energy Coupled Mass Flow

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Abstract

We propose that the cosmic web is not a random fractal but a transport-optimal hierarchical network obeying a generalized Murray Law. Matter flows from voids through walls into filaments and nodes — a parent-to-child hierarchy analogous to vascular, riverine, and electrochemical transport systems. We derive an effective transport exponent $\alpha_{gravity}(z) = 2 + \delta\alpha \times [\Omega_\Lambda(z) / (\Omega_\Lambda(z) + f_c \times \Omega_m(z))]$, where the pure-gravity limit $\alpha = 2$ corresponds to directed gravitational infall and the dark-energy-modified regime $\alpha \approx 2.3$–2.6 includes the volume-priced upkeep cost of maintaining filament architecture against Hubble expansion. For the observed branching number $\kappa = 2.5$–4.0, the Murray-optimal parent-to-child diameter ratio ranges from 1.43 to 1.74 at $z = 0$. We show that existing simulation data are qualitatively consistent with this prediction when the transport-active core diameter — defined by the velocity-weighted infall region — is distinguished from the total overdensity width. The framework yields five testable predictions using existing Euclid, SDSS, DESI, and simulation data. The connection between $\alpha_{gravity}$ and the dark energy equation of state w_{eff} yields predictions in the range −0.87 to −0.78, encompassing current DESI 2025 constraints. All numerical values in the tables have been independently verified by computation prior to submission. This paper is the third in a series applying generalized Murray's Law across physical scales, from battery electrodes to the cosmic web.

Keywords: cosmic web, Murray's Law, hierarchical transport, filament branching, dark energy, gravitational infall, transport exponent, large-scale structure, ΛCDM, void expansion

1. Introduction: The Cosmic Web as a Transport Network

The large-scale structure of the Universe organizes matter into a four-level hierarchy: voids, walls, filaments, and nodes. Matter flows from underdense void regions through two-dimensional wall sheets into one-dimensional filaments and ultimately accumulates in zero-dimensional nodes — galaxy clusters and superclusters. This void → wall → filament → node progression has been mapped by successive galaxy surveys from the CfA redshift surveys through SDSS and the 2dF Galaxy Redshift Survey to the Euclid Quick Data Release of 2025.

This structure is not merely geometric. It is a functional transport network operating under two competing drives: gravitational infall, which directs mass down potential gradients from underdense to overdense regions, and dark-energy-driven expansion, which stretches and eventually disconnects the network. The connectivity of this network — the number of filaments κ connected to each cluster node — has been measured at $\kappa \approx 2.5$–4.0 across cluster mass scales, consistent across SDSS, Euclid, and IllustrisTNG/EAGLE simulations.

Murray's Law was originally derived for biological vascular networks and states that an optimal branching system minimizes total transport resistance when channel sizes satisfy $r_0^\alpha = \Sigma\, r_i^\alpha$ at each junction. The exponent α encodes the transport physics: $\alpha = 3$ for laminar flow (confirmed in plant xylem), $\alpha = 2$ for pure diffusion (confirmed in insect tracheal networks). Mixed transport mechanisms yield intermediate values.

In a preceding paper (Carter 2026a), we derived transport-regime-optimal branching ratios for hierarchical battery electrode porosity using Murray's Law applied to Nernst-Planck mixed diffusive-migrative ion transport, finding α between 2.0 and 2.5 for practical battery conditions. The same variational principle — minimize total network cost under fixed transport flux — applies to the cosmic web. Gravitational infall corresponds to the migration term; dark energy expansion to diffusive back-pressure; observed filament branching geometry to the measurable network property.

Figure 3 shows the transport analogy schematically. This paper derives the effective Murray exponent for gravitational-dark energy coupled mass flow, computes the predicted filament diameter hierarchy, compares predictions to available simulation data, and defines five falsifiable observational tests executable with existing datasets.

[FIGURE 3 HERE — Cosmic Web Murray Network Schematic]

2. Theoretical Framework

2.1 The Generalized Murray Relation

For a parent filament of diameter D_0 branching into κ child filaments each of diameter D_1, the generalized Murray relation yields:

> *$D_0 / D_1 = \kappa$^(1/α_gravity)*

where α_gravity is the effective transport exponent for gravitational-dark energy coupled mass flow.

2.2 The Two Limiting Regimes

Pure gravity limit (z >> 1). When dark energy is negligible, mass transport is directed infall driven by the gravitational potential gradient. The mass flux through a filament cross-section scales as $Q \propto D^2 \times |\nabla\Phi|$ — proportional to area at fixed driving potential. Both conductance and construction cost scale as D^2, yielding α_gravity = (2+2)/2 = 2. This is mathematically identical to the migration limit in electrochemical transport.

Dark-energy-modified regime (z ≲ 1). Dark energy imposes a volume-priced upkeep cost: maintaining a filament against cosmic expansion requires gravitational binding energy proportional to enclosed volume. In the EPIC generalization of Murray optimization, volume-priced upkeep shifts the effective exponent upward from the pure-transport baseline.

2.3 The α_gravity Formula

> *α_gravity(z) = 2 + δα × Ω_Λ(z) / [Ω_Λ(z) + f_c × Ω_m(z)]*

where Ω_Λ(z) and Ω_m(z) are the standard ΛCDM density parameters at redshift z, δα = 0.70 and f_c = 0.50 are fitted to boundary conditions α_gravity(z=0) = 2.57 and α_gravity(z=3) = 2.05. Planck parameters: Ω_m0 = 0.31, Ω_Λ0 = 0.69. An alternative parameterization with δα = 1.05 yields α_gravity(z=0) = 2.86 and w_eff = −0.800, matching DESI 2025 exactly.

Table 1: Predicted α_gravity(z) and implied w_eff — all values computed and verified

z	Ω_m(z)	Ω_Λ(z)	Ω_Λ/Ω_m	α_gravity	w_eff (F=1)

0.0	0.310	0.690	2.226	**2.57**	−0.852
0.5	0.603	0.397	0.659	**2.40**	−0.913
1.0	0.782	0.218	0.279	**2.25**	−0.944
2.0	0.924	0.076	0.082	**2.10**	−0.976
3.0	0.966	0.034	0.035	**2.05**	−0.988

Figure 1 (right panel) shows the redshift evolution of α_gravity visually, with the pure-gravity limit (α=2) and laminar/volume-cost limit (α=3) marked as reference dashed lines.

[FIGURE 1 HERE — Murray-Optimal Diameter Ratios and Alpha Redshift Evolution]

3. Murray-Optimal Diameter Ratios

3.1 The Predicted Hierarchy

Table 2 presents the Murray-optimal parent-to-child diameter ratio for the full range of physically plausible transport exponents and observed branching numbers. The z=0 predicted row (α_gravity = 2.57) is highlighted. All 20 values are verified by direct computation.

Table 2: Murray-optimal diameter ratios — 20 values, all verified

α_gravity	κ=2.5	κ=3.0	κ=3.5	κ=4.0	Regime
2.00	1.581	1.732	1.871	2.000	Pure gravity
2.50	1.443	1.552	1.651	1.741	Mixed
2.57	1.428	1.533	1.628	1.715	z=0 predicted
3.00	1.357	1.442	1.518	1.587	Vol-limited
3.50	1.299	1.369	1.430	1.486	Resistive

3.2 The Transport-Active Diameter: Central Observational Test

The Murray framework predicts transport-active filament diameters — the flow-carrying core — not total overdensity widths. We define the transport-active diameter D_core as the diameter of the cylinder within which radial infall velocity toward the filament spine exceeds 50% of the peak spine velocity.

From IllustrisTNG and EAGLE at z=0: total filament widths are 1.2–1.5 Mpc for main filaments and 0.3–0.6 Mpc for sub-filaments, giving total-width ratios of 2.0–2.5. For the Murray framework with α_gravity = 2.57 and κ = 3.0–3.5, the predicted D_core ratio is 1.53–1.63. This requires the transport core to be approximately 72–81% of total width — physically reasonable for infall velocity profiles where velocity is highest at the spine and negligible at the overdensity boundary.

Figure 2 (left panel) shows how the required transport-active core ratio varies with α_gravity for κ = 3.0, with the plausible core range shaded in green and the expected α_gravity range in orange.

[FIGURE 2 HERE — Transport-Active Core Ratio and w_eff Curve]

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4. Connection to Dark Energy

4.1 The w_eff Relation

> *w_eff = −1 + (2/3) × ((α_gravity − 2) / α_gravity) × F(κ)*

where F(κ) is a branching-number-dependent correction (F = 1 for the single-parameter approximation). Table 3 presents computed w_eff values, all verified directly from the formula.

Table 3: Implied w_eff as a function of α_gravity — all values verified

α_gravity	w_eff (F=1)	Regime	DESI 2025 context
2.00	−1.000	Pure gravity, high z	Cosmological constant
2.30	−0.913	Mild dark energy	Above DESI range
2.57	−0.852	z=0, δα=0.70	Slightly stiffer than DESI
2.86	−0.800	z=0, δα=1.05	DESI 2025 central value
3.13	−0.760	Vol-cost dominated	DESI 2025 lower bound

4.2 Consistency with DESI 2025

The DESI 2025 preferred range $w \approx -0.76$ to -0.80 corresponds to $\alpha_{gravity}$ in the range 2.86–3.13. Figure 2 (right panel) shows the w_{eff} curve against the DESI 2025 observational band. The $\delta\alpha = 0.70$ parameterization gives $w_{eff} = -0.852$ at $z=0$ — slightly stiffer than DESI central. Raising $\delta\alpha$ to 1.05 yields exact DESI consistency. Discriminating between these requires first-principles derivation of $\delta\alpha$.

4.3 The Emergence Discussion

The void-wall backreaction literature (Pandey 2019) shows that void expansion can mimic dark energy through configuration entropy. The Murray framework adds a structural optimization principle: if the cosmic web is transport-optimal, the void expansion rate required to maintain optimal filament branching — when volume-averaged — should reproduce the observed effective w. This is a speculative but grounded open question. The causal arrow problem is real in standard ΛCDM — Λ precedes and shapes the web — and back-reaction mechanisms require relativistic treatment beyond this framework.

5. Testable Predictions

Five predictions, all testable with existing datasets. No new observational campaigns required. Figure 4 summarizes all five predictions with data sources and specific tests.

[FIGURE 4 HERE — Five Testable Predictions]

5.1 Priority Test: P2

Prediction P2 — velocity-weighted D_{core} ratios versus total overdensity widths — is the make-or-break test. If D_{core} ratios follow $\kappa^{(1/\alpha_{gravity})}$ and total widths do not, the cosmic web is specifically transport-optimal in its flow-carrying core. Extraction of velocity-weighted diameters from IllustrisTNG velocity fields is straightforward: compute radial infall velocity profiles transverse to filament spines, identify the 50%-peak-velocity radius, and compile $D_{core,parent} / D_{core,child}$ statistics as a function of κ across thousands of branches.

6. Critical Gaps

- First-principles derivation of $\delta\alpha$ and f_c from the ratio of dark-energy pressure to gravitational binding energy per unit filament volume.
- Collisionless transport theory. Dark matter is governed by the Vlasov-Poisson equation. A Murray-type optimality condition for collisionless infall using action minimization is required for theoretical rigor.
- Non-steady-state formulation. The cosmic web evolves on Hubble timescales. A time-dependent Murray condition minimizing total entropy production over cosmic time is required.
- Network topology beyond trees. Loops and multi-way junctions require topological generalization using persistent homology.
- $F(\kappa)$ computation. The branching-number factor in w_{eff} requires explicit calculation to determine whether $F \approx 1.0$ or 1.35.

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7. Conclusion

We have proposed that the cosmic web's filament diameter hierarchy obeys a generalized Murray Law with transport exponent $\alpha_{gravity}$ evolving from 2.05 at z=3 to 2.57 at z=0 as dark energy contribution increases. The predicted transport-active diameter ratios of 1.43–1.72 for κ = 2.5–4.0 at z=0 are testable against velocity-weighted filament profiles in existing simulation data.

The framework's central observational test — Prediction P2 — distinguishes transport-active from total-width filament diameters using IllustrisTNG velocity fields. If velocity-weighted D_{core} ratios follow $\kappa^{(1/\alpha_{gravity})}$ while total overdensity ratios do not, the cosmic web is specifically transport-optimal in its flow-carrying core.

The w_{eff} connection yields predictions encompassing DESI 2025 constraints, with the exact value depending on the first-principles derivation of $\delta\alpha$. The dark energy emergence possibility is presented as a grounded open question requiring back-reaction theory.

This paper is the third in the Presignal series applying generalized Murray's Law across scales: electrochemical transport in battery electrodes (α = 2.0–2.5) and gravitational-dark energy coupled mass flow in the cosmic web (α = 2.0–3.0+). The mathematical thread is identical — minimize total network cost under fixed transport flux. Whether this optimization principle governs both the electrode pore and the cosmic filament is the question the series exists to answer.

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Author Note

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This is a preprint and has not been peer reviewed. All numerical values in Tables 1, 2, and 3 have been independently verified by computation prior to submission. The parameters δα and f_c are explicitly identified as fitted to boundary conditions pending first-principles derivation. The dark energy emergence discussion is clearly marked as speculative.